PAN-ATLANTIC UNIVERSITY

School of Science and Technology

COURSE CODE: DTS304 **(STREAM 2)**
COURSE TITLE: Database Management I
EXAMINATION: Test (Written C. A.)
INSTRUCTION: Answer all questions **TIME:** 45 Minutes

Read the Case Study carefully and use it to answer Question 1

Case Study: Miniature University Database

Background:

The Database designers of a University Management System have been tasked to design a database based on the following **business rule**:

A faculty (a teaching staff) may belong to a department or be part of several departments, and a department must have at least one faculty.

Exactly one of the faculty in a department can head the department. A faculty cannot head more than a department.

Also, a department may offer many courses. Conversely, a course must be domiciled in exactly one department.

A student may take some courses and a course can have any number of students offering it. In this relationship, the grade a student got in a course must be recorded.

There are some additional information (the unique identifier in bold font):

1. A student has the following attributes: **ssn**, firstname, lastname, and dob.
2. A course has the following attributes: **courseid**, name, credits, and description.
3. A faculty is identified by **facultyid**, firstname, lastname, and rank.
4. A department is identified by the following: **deptID** and name.

Question 1

Draw the detailed entity relationship diagram (ERD) for the ultimate pet shop business entity. Pay attention to the following: **[12.5 marks]**

- The attributes of the entities must be shown and the unique identifier for each entity must be depicted.
- Show the minimum and maximum cardinality constraints for each entity participating in a relationship.

Question 2

(a) Explain the following concepts with respect to database modelling using a simple case study/scenario.

i. Super Keys **[1.5 marks]**

ii. Candidate Keys **[1.5 marks]**

iii. Foreign Keys **[1.5 marks]**

(b) Explain three fundamental building blocks of a business model. **[3 marks]**